

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8	(a)			More efficient/ efficiency has increased (1) Any one (×1) from: • as food production has increased, the amount of land used (for food production) has remained almost the same (1) ACCEPT use of data, e.g. land use increase 15% and food production increased 275% • Over time more food is being produced {per unit area of land/ on less land} (1) • yield {has increased/is better} (1) • (because of) intensive farming (1)			2	2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			Indicative content: A. Plant {leguminous plants / ref. clover, peas, beans/ plants which have nitrogen fixing bacteria} B. (Nitrogen fixing bacteria fix nitrogen that is) used to make {protein/ amino acids/ nitrates} / plants use nitrates to make {proteins/ amino acids} C. {Spread/use} {manure/ slurry/ animal waste/ compost} contains {protein/ urea} / plough in plants which contain protein D. this is {converted/turned into/ broken down/ decayed} to ammonia E. by {bacteria/ fungi/ decomposers}/ correct ref to urease F. ammonia is converted to nitrates G. Use {fertilisers (containing nitrates) / NPK}/ add nitrates H. nitrates {taken up/ absorbed/ used} by plants I. {draining / ploughing} prevents {denitrification / denitrifying bacteria} / {draining / ploughing} promotes {nitrification / nitrifying bacteria / nitrogen fixing bacteria}	3	3		6		

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				5-6 marks At least 7 points from indicative content <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3-4 marks At least 4 points from indicative content <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks At least 1 point from indicative content <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
Question 8					3	3	2	8	0	0